

REPORT

LOAN PORTFOLIO PERFORMANCE ANALYSIS

1. Executive Summary

This project aims to analyze the performance of a bank's loan portfolio using SQL and Power BI. The analysis focuses on evaluating lending activities, repayment performance, portfolio quality, borrower characteristics, and risk factors affecting loan defaults.

The report was developed based on historical loan application data and includes data cleaning, exploratory data analysis (EDA), and interactive dashboard development. The final solution provides management with a comprehensive overview of loan operations, identifies key risk segments, and supports data-driven lending decisions.

Key findings indicate that the loan portfolio is generally healthy, with Good Loans accounting for 86.2% of total applications. However, certain borrower segments, including lower-grade loans, renters, and borrowers with limited employment history, exhibit significantly higher default risk.

2. Data Preparation

2.1. Dataset Overview

The dataset contains information related to loan applications, borrower profiles, loan characteristics, repayment activities, and loan performance.

Key variables include:

- Loan Amount
- Total Payment
- Interest Rate
- Debt-to-Income Ratio (DTI)
- Grade and Sub-grade
- Loan Status
- Home Ownership
- Employment Length

- Annual Income
- Loan Purpose
- Term

2.2. Data Cleaning

Before conducting the analysis, several data quality checks were performed:

- Checked for duplicate Loan IDs.
- Examined missing values across all columns.
- Validated loan status categories.
- Converted date fields into standard date formats.
- Verified numerical fields for consistency and accuracy.

The dataset was deemed suitable for further analysis after completing the preprocessing stage.

3. Exploratory Data Analysis (EDA)

3.1. Loan Status Distribution

The portfolio consists primarily of Fully Paid loans, while Charged Off loans represent a smaller but important portion of the portfolio.

Observation

- Fully Paid loans dominate the portfolio.
- Charged Off loans account for approximately 14% of total applications.
- Portfolio quality remains relatively healthy.

3.2. Default Rate by Loan Grade

The analysis reveals a clear relationship between loan grade and default risk.

Grade	Default Rate
A	5.26%

Grade	Default Rate
B	19.72%
C	25.81%
D	21.43%
E	0.00%

Insight

Default rates increase significantly from Grade A to Grades B–D. This suggests that the grading system effectively captures borrower risk and can be used as a reliable indicator for credit assessment.

3.3. Default Rate by Home Ownership

Grade	Default Rate
RENT	20.21%
OWN	10.17%
MORTGAGE	6.09%

Insight

Borrowers who rent their homes exhibit substantially higher default rates compared to homeowners and borrowers with mortgages, suggesting lower financial stability within this segment.

3.4. Default Rate by Employment Length

Borrowers with less than one year of employment experience recorded the highest default rate among all employment groups.

Insight

Employment stability appears to play an important role in repayment behavior. Longer employment history is generally associated with lower credit risk.

3.5. Average Annual Income by Grade

Grade	Average Income
A	\$63,783
B	\$64,965
C	\$60,536
D	\$61,021
E	\$60,000

Insight

Average income remains relatively consistent across loan grades. This suggests that loan grading is influenced by multiple factors beyond income alone, such as credit history and repayment behavior.

3.6. Average Interest Rate by Grade

Grade	Average Interest Rate
A	7.01%
B	11.01%
C	13.48%
D	15.32%
E	16.29%

Insight

Interest rates increase steadily with risk level, demonstrating the bank's risk-based pricing strategy.

3.7. Loan Term Analysis

Term	Average Funded Amount
36 Months	\$5,985
60 Months	\$3,708

Insight

Borrowers tend to obtain larger loans under shorter-term agreements, indicating stronger demand for shorter repayment periods.

3.8. Monthly Lending Trend

Loan applications remain relatively stable throughout the year, while funded amounts increase noticeably toward year-end.

Insight

December records the highest funded amount, suggesting stronger lending activity during the final quarter of the year.

4. SQL Analysis

4.1. BANK LOAN REPORT | SUMMARY

KPI's:

Total Loan Applications

```
SELECT COUNT(id) AS Total_Applications
FROM bank_loan_data
```

	Total_Applications
1	38576

The bank processed 38.6K loan applications during the reporting period, indicating strong lending demand and a sizable customer base.

MTD Loan Applications

```
SELECT COUNT(id) AS Total_Applications
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	Total_Applications
1	4314

PMTD Loan Applications

```
SELECT COUNT(id) AS Total_Applications
FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	Total_Applications
1	4035

Loan applications increased by approximately 6.9% compared to the previous month, reflecting growing borrowing activity.

Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount
FROM bank_loan_data
```

	Total_Funded_Amount
1	435757075

The bank disbursed over \$435M in loans, demonstrating significant lending volume and market presence.

MTD Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	Total_Funded_Amount
1	53981425

PMTD Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	Total_Funded_Amount
1	47754825

Funded amount grew by around 13.0% month-over-month, suggesting stronger loan issuance in December.

Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected
FROM bank_loan_data
```

	Total_Amount_Collected
1	473070933

Total repayments exceeded funded amounts, indicating healthy cash inflows and portfolio performance.

MTD Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected
FROM bank_loan_data
```

WHERE MONTH(issue_date) = 12

	Total_Amount_Collected
1	58074380

PMTD Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected
FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	Total_Amount_Collected
1	50132030

Collections increased by 15.8% from the previous month, reflecting improved repayment performance.

Average Interest Rate

```
SELECT AVG(int_rate)*100 AS Avg_Int_Rate
FROM bank_loan_data
```

	Avg_Int_Rate
1	12.0488314172048

The portfolio maintains an average lending rate of approximately 12%, balancing profitability and borrower affordability.

MTD Average Interest

```
SELECT AVG(int_rate)*100 AS MTD_Avg_Int_Rate
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	MTD_Avg_Int_Rate
1	12.3560408676042

PMTD Average Interest

```
SELECT AVG(int_rate)*100 AS PMTD_Avg_Int_Rate
FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	PMTD_Avg_Int_Rate
1	11.9417175498261

Average interest rates increased slightly in December, indicating a shift toward higher-risk or higher-yield loans.

Avg DTI

```
SELECT AVG(dti)*100 AS Avg_DTI
FROM bank_loan_data
```

	Avg_DTI
1	13.3274331211432

Borrowers generally maintain a manageable debt-to-income ratio, suggesting acceptable repayment capacity.

MTD Avg DTI

```
SELECT AVG(dti)*100 AS MTD_Avg_DTI
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	MTD_Avg_DTI
1	13.6655377880425

PMTD Avg DTI

```
SELECT AVG(dti)*100 AS PMTD_Avg_DTI
FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	PMTD_Avg_DTI
1	13.3027335836364

DTI increased marginally, implying slightly higher borrower leverage in recent applications.

GOOD LOAN ISSUED

Good Loan Percentage

```
SELECT
  (COUNT(CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'Current' THEN id END) *
  100.0) /
  COUNT(id) AS Good_Loan_Percentage
FROM bank_loan_data
```

	Good_Loan_Percentage
1	86.175342181667

The vast majority of loans are classified as Good Loans, indicating a healthy and well-performing loan portfolio.

Good Loan Applications

```
SELECT COUNT(id) AS Good_Loan_Applications
FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Applications
1	33243

Good Loan Funded Amount


```
SELECT SUM(loan_amount) AS Good_Loan_Funded_amount
FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Funded_amount
1	370224850

Good Loan Amount Received

```
SELECT SUM(total_payment) AS Good_Loan_amount_received
FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_amount_received
1	435786170

Good Loans account for most applications, funding, and repayments, highlighting strong portfolio quality and repayment behavior.

BAD LOAN ISSUED

Bad Loan Percentage

```
SELECT
    (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0) /
    COUNT(id) AS Bad_Loan_Percentage
FROM bank_loan_data
```

	Bad_Loan_Percentage
1	13.824657818332

Approximately 14% of loans were charged off, representing the primary source of credit risk within the portfolio.

Bad Loan Applications

```
SELECT COUNT(id) AS Bad_Loan_Applications
FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	Bad_Loan_Applications
1	5333

Bad Loan Funded Amount

```
SELECT SUM(loan_amount) AS Bad_Loan_Funded_amount
FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	Bad_Loan_Funded_amount
1	65532225

Bad Loan Amount Received

```
SELECT SUM(total_payment) AS Bad_Loan_amount_received
FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	Bad_Loan_amount_received
1	37284763

Bad Loans generated substantially lower repayment amounts compared to funding provided, resulting in financial losses for the bank.

LOAN STATUS

```
SELECT
    loan_status,
    COUNT(id) AS LoanCount,
    SUM(total_payment) AS Total_Amount_Received,
    SUM(loan_amount) AS Total_Funded_Amount,
    AVG(int_rate * 100) AS Interest_Rate,
    AVG(dti * 100) AS DTI
FROM bank_loan_data
GROUP BY loan_status
```

	loan_status	LoanCount	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
1	Fully Paid	32145	411586256	351358350	11.6410707918092	13.1673507557434
2	Current	1098	24199914	18866500	15.0993260800947	14.7243442736843
3	Charged Off	5333	37284763	65532225	13.8785749318289	14.0047328005517

```
SELECT
    loan_status,
    SUM(total_payment) AS MTD_Total_Amount_Received,
    SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
GROUP BY loan_status
```

	loan_status	MTD_Total_Amount_Received	MTD_Total_Funded_Amount
1	Fully Paid	47815851	41302025
2	Current	4934318	3946625
3	Charged Off	5324211	8732775

Fully Paid loans dominate the portfolio and generate the highest repayment value, indicating strong portfolio performance.

Charged-Off loans exhibit higher interest rates and DTI levels, suggesting increased borrower risk.

Current loans carry the highest average interest rates and DTI levels, reflecting ongoing exposure to higher-risk borrowers.

4.2. BANK LOAN REPORT | OVERVIEW

MONTH

```
SELECT
```

```

MONTH(issue_date) AS Month_Munber,
DATENAME(MONTH, issue_date) AS Month_name,
COUNT(id) AS Total_Loan_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)
ORDER BY MONTH(issue_date)

```

	Month_Munber	Month_name	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

Loan applications, funded amounts, and repayments increased steadily throughout the year. December recorded the highest values across all three metrics, indicating strong year-end lending demand and repayment activity.

STATE

```

SELECT
    address_state AS State,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY address_state
ORDER BY address_state

```

	State	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received					
1	AK	78	1031800	1108570	26	MS	19	139125	149342
2	AL	432	4949225	5492272	27	MT	79	829525	892047
3	AR	236	2529700	2777875	28	NC	759	8787575	9534813
4	AZ	833	9206000	10041986	29	NE	5	31700	24542
5	CA	6894	78484125	83901234	30	NH	161	1917900	2101386
6	CO	770	8976000	9845810	31	NJ	1822	21657475	23425159
7	CT	730	8435575	9357612	32	NM	183	1916775	2084485
8	DC	214	2652350	2921854	33	NV	482	5307375	5451443
9	DE	110	1138100	1269136	34	NY	3701	42077050	46108181
10	FL	2773	30046125	31601905	35	OH	1188	12991375	14330148
11	GA	1355	15480325	16728040	36	OK	293	3365725	3712649
12	HI	170	1850525	2080184	37	OR	436	4720150	4966903
13	IA	5	56450	64482	38	PA	1482	15826525	17462908
14	ID	6	59750	65329	39	RI	196	1883025	2001774
15	IL	1486	17124225	18875941	40	SC	464	5080475	5462458
16	IN	9	86225	85521	41	SD	63	606150	656514
17	KS	260	2872325	3247394	42	TN	17	162175	141522
18	KY	320	3504100	3792530	43	TX	2664	31236650	34392715
19	LA	426	4498900	5001160	44	UT	252	2849225	2952412
20	MA	1310	15051000	16676279	45	VA	1375	15982650	17711443
21	MD	1027	11911400	12985170	46	VT	54	504100	534973
22	ME	3	9200	10808	47	WA	805	8855525	9531739
23	MI	685	7829900	8543660	48	WI	446	5070450	5485161
24	MN	592	6302600	6750746	49	WV	167	1830525	1991936
25	MO	660	7151175	7692732	50	WY	79	890750	1046050

California (CA) and New York (NY) contribute the highest number of loan applications and funding volumes, making them the bank's most important lending markets.

TERM

SELECT

```
term AS Term,
COUNT(id) AS Total_Loan_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
SUM(total_payment) AS Total_Amount_Received
```

FROM bank_loan_data

GROUP BY term

ORDER BY term

	Term	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

36-month loans are significantly more popular than 60-month loans, accounting for nearly three-quarters of all applications.

EMPLOYEE LENGTH

SELECT

```
emp_length AS Employee_Length,
COUNT(id) AS Total_Loan_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
SUM(total_payment) AS Total_Amount_Received
```

FROM bank_loan_data

GROUP BY emp_length

ORDER BY emp_length

	Employee_Length	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	< 1 year	4575	44210625	47545011
2	1 year	3229	32883125	35498348
3	10+ years	8870	116115950	125871616
4	2 years	4382	44967975	49206961
5	3 years	4088	43937850	47551832
6	4 years	3428	37600375	40964850
7	5 years	3273	36973625	40397571
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

Borrowers with more than 10 years of employment represent the largest customer segment, generating the highest application volume, funding amount, and repayments. Longer employment history appears positively associated with loan demand.

PURPOSE

```
SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY purpose
ORDER BY purpose
```

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	car	1497	10223575	11324914
2	credit card	4998	58885175	65214084
3	Debt consolidation	18214	232459675	253801871
4	educational	315	2161650	2248380
5	home improvement	2876	33350775	36380930
6	house	366	4824925	5185538
7	major purchase	2110	17251600	18676927
8	medical	667	5533225	5851372
9	moving	559	3748125	3999899
10	other	3824	31155750	33289676
11	renewable_energy	94	845750	898931
12	small business	1776	24123100	23814817
13	vacation	352	1967950	2116738
14	wedding	928	9225800	10266856

Debt Consolidation is the dominant borrowing purpose, accounting for nearly half of total applications and funding volume.

Credit card refinancing is the second most common reason for borrowing, indicating strong consumer demand for debt management solutions.

HOME OWNERSHIP

```
SELECT
    home_ownership AS Home_Ownership,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY home_ownership
ORDER BY home_ownership
```

	Home_Ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	MORTGAGE	17198	219329150	238474438
2	NONE	3	16800	19053
3	OTHER	98	1044975	1025257
4	OWN	2838	29597675	31729129
5	RENT	18439	185768475	201823056

Most borrowers either Rent or have a Mortgage, accounting for nearly all loan applications. Mortgage borrowers receive the highest funded amount, while Rent borrowers represent the largest customer group. Borrowers with Own, Other, or None home ownership contribute only a small share of the portfolio.

```
SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
WHERE grade = 'A'
GROUP BY purpose
ORDER BY purpose
```

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	car	577	3629475	3805538
2	credit card	1353	12636075	13339495
3	Debt consolidation	3753	37216300	38822971
4	educational	79	484000	515639
5	home improvement	933	8359175	8744006
6	house	91	916575	957878
7	major purchase	796	5344575	5604259
8	medical	197	1471850	1526882
9	moving	164	1069450	1099875
10	other	1024	7043175	7397982
11	renewable_energy	29	224150	225827
12	small business	334	3172075	3190467
13	vacation	122	683625	694542
14	wedding	237	2001725	2126202

Borrowers with Grade A maintain a well-diversified loan portfolio, with Debt Consolidation remaining the primary borrowing purpose. Lower-risk borrowers also show relatively high demand for Credit Card refinancing and Home Improvement loans.

```

SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
WHERE grade = 'B'
GROUP BY purpose
ORDER BY purpose

```

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	car	482	3494775	3885379
2	credit card	1603	18282500	20105335
3	Debt consolidation	5504	68427300	74088639
4	educational	98	701900	688114
5	home improvement	869	10596875	11439486
6	house	104	1406900	1471332
7	major purchase	588	5363225	5783676
8	medical	197	1489625	1595223
9	moving	176	1228425	1310297
10	other	1160	9624375	10266162
11	renewable_energy	32	298300	342066
12	small business	463	6321400	6073268
13	vacation	119	712375	772706
14	wedding	279	2756000	2953332

Grade B follows a similar borrowing pattern to Grade A. Debt Consolidation dominates loan demand, while Credit Card and Home Improvement remain the second and third largest purposes.

```

SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(Loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
WHERE grade = 'C'
GROUP BY purpose
ORDER BY purpose

```

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	car	241	1571650	1778019
2	credit card	1052	12403025	13834371
3	Debt consolidation	3868	47588125	52563059
4	educational	90	701425	767451
5	home improvement	520	6117650	6731732
6	house	70	945825	1029482
7	major purchase	372	2857950	3179844
8	medical	133	1142750	1167080
9	moving	112	667175	719889
10	other	827	6268250	6620343
11	renewable_energy	14	101700	96619
12	small business	346	4855375	4927528
13	vacation	60	266900	299126
14	wedding	199	1968650	2258975

As borrower risk increases, Debt Consolidation continues to be the dominant purpose. Credit Card refinancing remains the second largest category, suggesting that debt management is a common financial need among medium-risk borrowers.

```
SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(Loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
WHERE grade = 'D'
GROUP BY purpose
ORDER BY purpose
```

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	car	128	951800	1128398
2	credit card	582	7999325	9047851
3	Debt consolidation	2771	37007125	41314725
4	educational	30	168225	171234
5	home improvement	295	3785800	4205409
6	house	50	707725	809769
7	major purchase	224	1954775	2108792
8	medical	83	687175	751709
9	moving	65	387350	437787
10	other	491	4293200	4673952
11	renewable_energy	9	107700	128745
12	small business	292	4333450	4238232
13	vacation	28	151150	186244
14	wedding	134	1386000	1621044

Grade D borrowers exhibit a similar loan purpose distribution, although funded amounts become more concentrated in Debt Consolidation loans. Small Business loans also represent a noticeable share compared to higher-grade borrowers.

```
SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(Loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
WHERE grade = 'E'
GROUP BY purpose
ORDER BY purpose
```

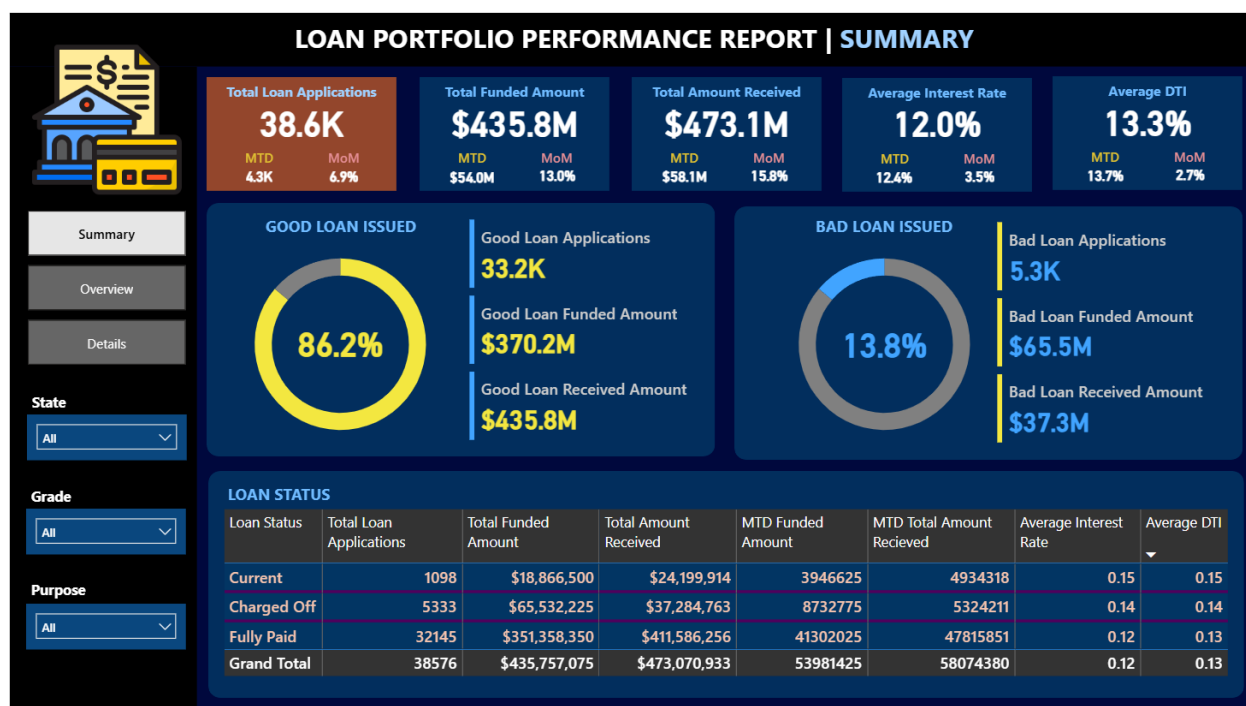
	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	car	56	435675	558818
2	credit card	279	4731175	5364826
3	Debt consolidation	1561	27033875	30205711
4	educational	13	63100	70300
5	home improvement	183	3058300	3573231
6	house	30	456650	470302
7	major purchase	98	1167700	1377960
8	medical	40	437800	505613
9	moving	28	248500	246550
10	other	220	2379550	2553313
11	renewable_energy	5	30650	43736
12	small business	200	3224400	3199825
13	vacation	20	129300	131319
14	wedding	53	768425	862647

Despite representing the highest credit risk, Grade E borrowers still primarily seek loans for Debt Consolidation. Credit Card refinancing remains the second largest purpose, indicating that financially stressed borrowers mainly borrow to restructure existing debt.

Overall Observation: Across all credit grades (A–E), Debt Consolidation consistently accounts for the largest number of applications and funded amount, followed by Credit Card loans. This suggests that managing existing debt is the primary reason customers apply for loans, regardless of their credit quality.

5. Dashboard Analysis

5.1. Dashboard 1 – Summary Analysis



Portfolio Overview

The loan portfolio contains **38.6K loan applications** with a total funded amount of **\$435.8M** and total amount received of **\$473.1M**.

The difference between total amount received and total funded amount indicates that the bank has generated additional revenue through interest payments. The overall average interest rate is **12.0%**, while the average Debt-to-Income (DTI) ratio is **13.3%**, suggesting that borrowers generally maintain a manageable debt burden relative to their income.

Month-to-Date (MTD) metrics show continued growth across all major indicators:

- Loan Applications reached **4.3K**, representing a **6.9% increase** compared to the previous month.
- Funded Amount increased by **13.0% MoM** to **\$54.0M**.
- Amount Received increased by **15.8% MoM** to **\$58.1M**.
- Average Interest Rate increased slightly to **12.4%**.
- Average DTI rose to **13.7%**.

These trends suggest that lending activity accelerated toward the end of the year while repayment collections remained strong.

Good Loan vs Bad Loan Performance

The portfolio is dominated by high-quality loans.

Good Loans

- Good Loan Applications: **33.2K**
- Good Loan Percentage: **86.2%**
- Funded Amount: **\$370.2M**
- Amount Received: **\$435.8M**

The high proportion of good loans indicates effective credit assessment and strong borrower repayment performance. Good loans account for the majority of both funded capital and cash inflows.

Bad Loans

- Bad Loan Applications: **5.3K**
- Bad Loan Percentage: **13.8%**
- Funded Amount: **\$65.5M**
- Amount Received: **\$37.3M**

Although bad loans represent a relatively small portion of total applications, they account for a significant amount of capital exposure. The lower amount received compared to funded amount highlights losses associated with loan defaults and charge-offs.

Loan Status Analysis

The loan portfolio consists of three major loan statuses:

Loan Status	Applications	Funded Amount
Fully Paid	32,145	\$351.4M

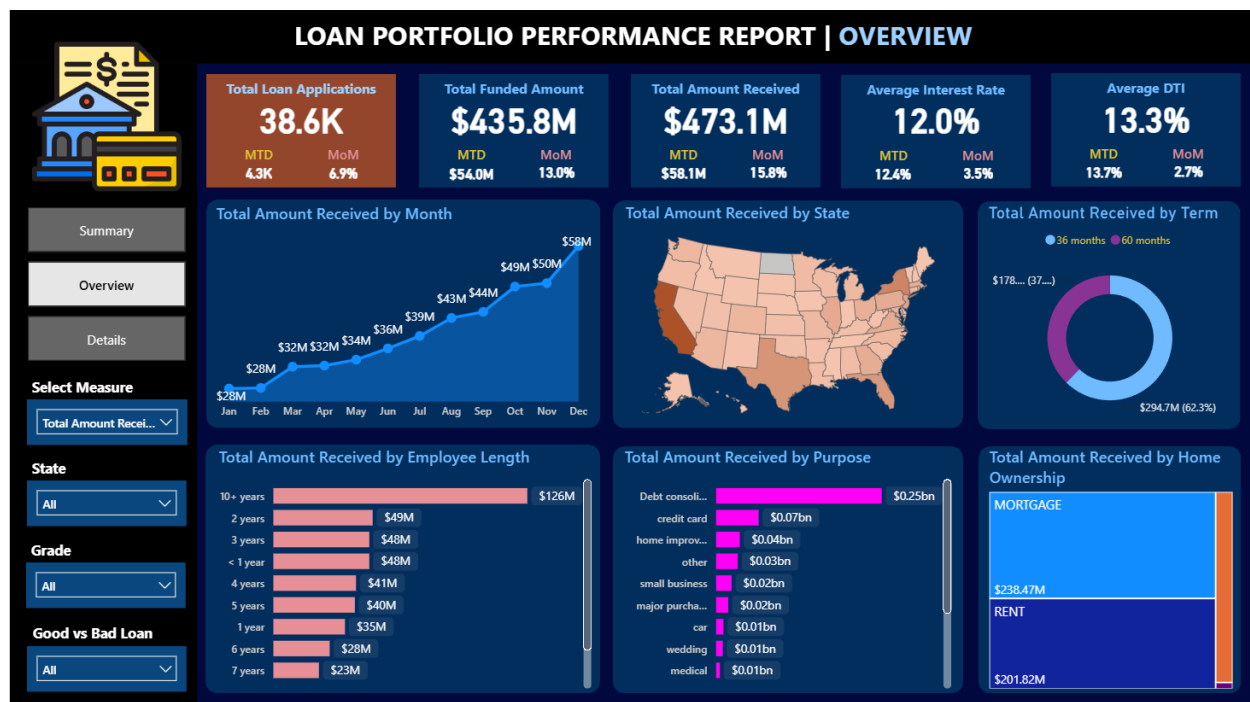
Loan Status	Applications	Funded Amount
Charged Off	5,333	\$65.5M
Current	1,098	\$18.9M

Key observations:

- **Fully Paid loans dominate the portfolio**, accounting for more than four-fifths of all applications.
- **Charged Off loans represent the primary source of credit risk**, with over \$65M in funded capital.
- **Current loans account for a relatively small portion**, indicating that most loans in the dataset have already reached their final repayment outcome.

The average interest rate for Charged Off loans is higher than Fully Paid loans, suggesting that higher-risk borrowers were generally charged higher borrowing costs.

5.2. Dashboard 2 – Overview Analysis



Monthly Lending Trends

Loan applications demonstrate a generally upward trend throughout the year.

Key findings:

- Applications increased from approximately **2.3K in January** to **4.3K in December**.
- Funded amount increased steadily, reaching its highest level of **\$54M in December**.
- Amount received also peaked in December at approximately **\$58M**.

This consistent growth indicates increasing demand for loan products and effective loan origination activities throughout the year.

The alignment between funded amount growth and amount received growth suggests that portfolio expansion did not negatively impact repayment performance.

Regional Analysis

State-level analysis reveals significant geographic concentration of lending activity.

Observations:

- A small number of states contribute a large proportion of total applications and funded amounts.
- California appears to be one of the largest lending markets.
- Lending activity is broadly distributed across the United States, indicating a diversified geographic portfolio.

Geographic diversification helps reduce concentration risk and protects the portfolio from regional economic downturns.

Loan Term Analysis

The portfolio is heavily concentrated in **36-month loans**.

Term	Share
36 Months	73.2%

Term	Share
60 Months	26.8%

Additional analysis shows:

- Average funded amount for 36-month loans is significantly higher than 60-month loans.
- Borrowers generally prefer shorter repayment periods.
- Shorter-term loans typically reduce long-term default risk and improve portfolio liquidity.

This distribution suggests a conservative lending strategy focused on shorter-duration loans.

Employment Length Analysis

Borrowers with longer employment histories account for the largest share of lending activity.

Key findings:

- Borrowers with **10+ years of employment** contribute the highest number of applications, funded amounts, and repayments.
- Borrowers with less than one year of employment also represent a sizable customer segment.
- Mid-career borrowers (2-6 years of employment) maintain consistent borrowing activity.

Longer employment history generally reflects greater income stability, which may contribute to improved repayment performance.

Loan Purpose Analysis

Loan demand is highly concentrated in a few major borrowing categories.

Major loan purposes include:

1. Debt Consolidation
2. Credit Card Refinancing
3. Home Improvement
4. Major Purchases
5. Small Business Financing

Debt Consolidation represents the largest loan purpose by a significant margin.

This suggests that many borrowers use loans primarily to restructure existing debt obligations rather than finance new expenditures.

Understanding loan purpose distribution helps identify customer financing needs and supports product development initiatives.

Home Ownership Analysis

Mortgage holders and renters represent the largest borrower groups.

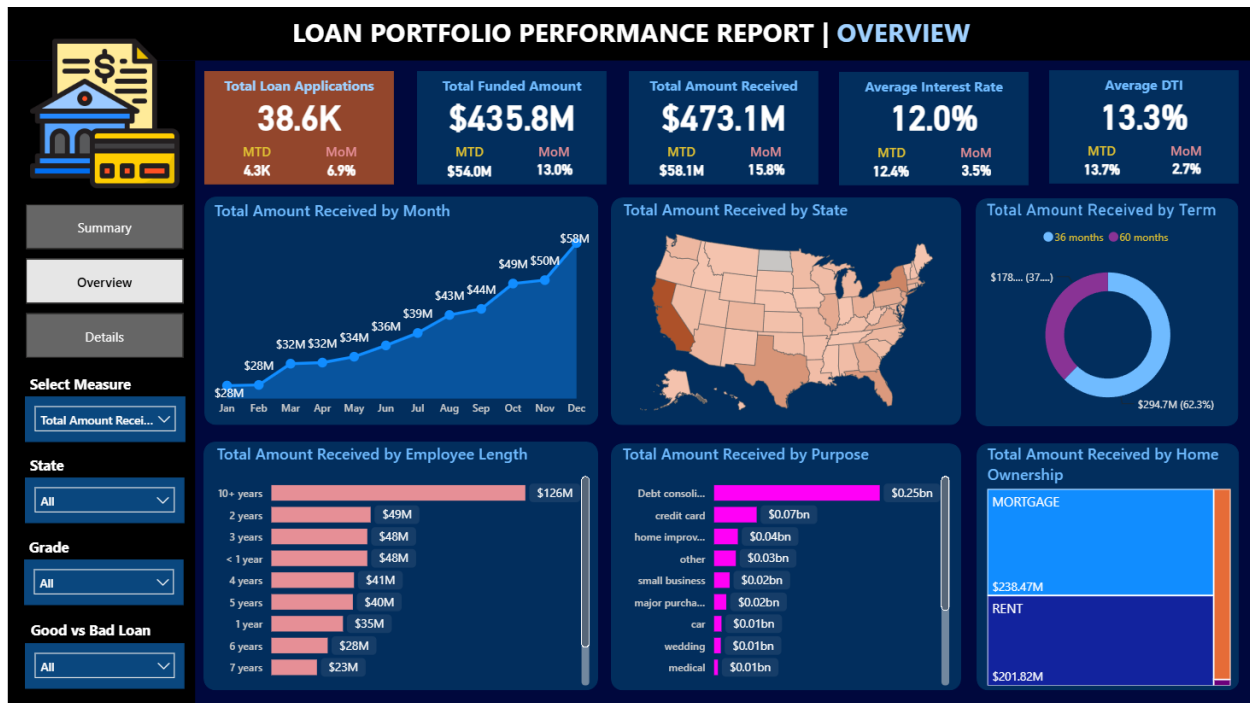
Key observations:

- Mortgage borrowers contribute the highest funded amount.
- Renters account for a similarly large share of loan applications.
- Homeowners contribute a smaller but still meaningful portion of the portfolio.

This distribution indicates that the bank serves a diverse customer base across different housing situations.

Mortgage holders may demonstrate stronger financial stability due to established asset ownership, while renters provide a large source of lending demand.

5.3. Dashboard 3 – Details Analysis



The Details Dashboard provides a transaction-level view of the loan portfolio and supports detailed operational analysis.

Users can drill down into individual loan records using multiple filters, including:

- State
- Grade
- Loan Status
- Purpose
- Selected Measure

Each record contains key borrower and loan attributes such as:

- Loan ID
- Purpose
- Home Ownership
- Grade and Sub-Grade
- Issue Date
- Funded Amount
- Interest Rate

- Installment Amount
- Amount Received

The dashboard enables users to:

- Investigate specific loans contributing to portfolio performance.
- Identify high-risk borrowers and loan segments.
- Validate trends observed in the Summary and Overview dashboards.
- Support operational monitoring and portfolio management activities.

The detailed data view enhances transparency and allows stakeholders to move seamlessly from high-level KPIs to individual loan-level records.

6. Overall Business Conclusion

The loan portfolio demonstrates **strong overall performance**, with **86.2% of applications classified as Good Loans** and more than **\$473M collected against \$436M funded**.

Lending activity increased steadily throughout the year, reaching peak levels in December. The portfolio is primarily composed of **36-month loans**, borrowers with **long employment histories**, and loans used for **debt consolidation purposes**.

While **Charged Off loans account for only 13.8% of applications**, they represent a meaningful source of credit risk and should remain a focus area for monitoring and risk management.

Overall, the portfolio appears well-balanced, geographically diversified, and supported by strong repayment performance, indicating effective lending and credit evaluation practices.